MILITARY SPECIFICATION

TRUCK, AIRCRAFT LUBRICATING OIL DISPENSING UNIT 200 GALLON, SELF-CONTAINED, SKID MOUNTED, 4 BY 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.2 Scope. This specification covers a commercial, gasoline-engine-driven, four wheel, two rear wheel drive, truck, mounted with a self-contained 200-gallon capacity, aircraft lubricating oil dispensing skid mounted unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-393 - Extinguisher, Fire, Dry-Chemical (Hard-Portable).
W-B-131 - Battery, Storage, Vehicular, Ignition Lighting, and Starting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2320

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MILITARY

MIL-L-22851 - Lubricating Oil, Aircraft Piston Engine (Ashless Dispersant).

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Automotive Vehicle.

FED-STD-307 - Trucks, Light Commercial, Gasoline/Diesel Fueled, Two-Wheel Driven (4 by 2), 3,000 to 11,000 pounds GVW.

FED-STD-595 - Colors.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.

MIL-STD-1223 - Non-Tactical Wheeled Vehicles; Treatment, Painting, Identification Marking and Data Plate Standards.

MIL-STD-1474 - Noise Limits for Army Materiel.

2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852, or CFR, Title 40 should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The aviation-lube (Av-lube) unit/truck shall be a self-contained, skid mounted, gasoline-engine-powered air compressor operating an air pump to dispense a single type aircraft lubricating oil from a 200 gallon aluminum alloy tank. The Av-lube unit/truck shall be equipped with

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an air pumping system, oil servicing nozzle, oil heating system and all the components required to make a complete aircraft lubricating oil dispensing system. The Av-lube unit/truck shall be capable of loading the 200 gallon tank from a drum on the ground. The Av-lube unit/truck shall be dimensioned to fit within the confines of an 8 foot pickup truck bed.

3.2 First production vehicle. When specified (see 6.2), the contractor shall furnish one complete Av-lube unit/truck for first production vehicle inspection (see 4.2.1 and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Maintainability. The Av-lube unit/truck design, together with components and accessory location and installation, shall permit ready accessability to all items requiring periodic maintenance service in the field, which will be accomplished with the use of conventional general purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum removal of the components.

3.5 Standard truck, components, and accessories. Except as specified in 3.5.1 through 3.5.5 the truck, components, and accessories shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specification and technical materials, identical to that furnished to the authorized company representative for selection of truck models and components. Truck, components, assemblies, and accessories to be provided hereunder are standard or optional commercial products for the purpose of this specification. Such items shall be construed as commercial motor vehicle products, and shall comply with, FED-STD-307, Type 4, Class F, and DoT Federal Motor Vehicle Safety Standards and Regulations applicable, on the date of manufacture.

3.5.1 Cooling system. Manufacturers standard cooling system shall be furnished that shall maintain engine coolant at a temperature below the boiling point with truck loaded to rated gross vehicle weight (gvw).

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3.5.2 Coolant system and indicators. Coolant system shall include a surge tank, a deaeration system, or shall include a coolant recovery reservoir of not less than 2 quart capacity.

3.5.3 Radio noise suppressors. The truck chassis shall be equipped with radio noise suppressors conforming to SAE J552. Electromagnetic radiation from the truck chassis shall be within the limits of SAE J551.

3.5.4 Sound level. The exterior sound level produced by the truck shall not exceed the noise limits established by the EPA in EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks. The interior sound level shall be in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.94.

3.5.5 Air pollution control. Trucks shall comply with the EPA regulations governing control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, trucks destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.6 Weights, ratings, and dimensions.

3.6.1 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricants and coolant.

3.6.2 Weight rating. The gvw rating shall consist of curb weight, operator (weight computed at 175 pounds (lbs)), and payload not to exceed 9,000 lb gvw.

3.6.3 Weight distribution. The distribution of the gvw for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distribution over the load area.

3.6.4 Ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. Minimum payload shall be 3,000 lbs.

3.6.5 Cab-to-axle dimension. Shall be compatible with the mounting of an 8-foot fleetside pickup truck box for the Av-lube unit, allowing normal clearances where necessary.

3.7 Performance.

3.7.1 Speeds and gradeability. Speed and gradeability requirements shall be met with the vehicle loaded to its minimum specified gvw.

3.7.1.1 Speed. The vehicle loaded to the specified gvw (see 3.6.2), and all specified equipment and devices installed and operable shall be capable of maintaining at sea level a continuous speed of not less than 50 miles per hour (mph) on a smooth hard-surfaced, level road.

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3.7.1.2 Gradeability. High and low speed gradeability requirements shall be met as specified herein. Vehicle shall ascend a continuous grade of 1.5 percent at 55 mph when fully loaded. High speed gradeability requirements shall be met with the transmission in drive.

3.8 Chassis components.

3.8.1 Engine. The truck engine shall be a liquid cooled, four-stroke cycle, 6 cylinder gasoline engine, having horsepower (hp), torque, and speed characteristics to satisfactorily meet all truck performance requirements specified herein. The engine furnished shall produce the required truck performance when operated on unleaded gasoline. The engine shall be capable of continuous, warranted operation on unleaded fuel.

3.8.1.1 Oil filter. Full flow type oil filter shall be furnished.

3.8.2 Electrical equipment.

3.8.2.1 Ignition system. The truck shall be equipped with a 12 volt (V) ignition system. An alternator of not less than 40 amperes rated capacity, and which provides not less than 16 amperes direct current output at normal engine idle speed, shall be furnished.

3.8.2.2 Lighting and wiring. The truck lighting and wiring shall conform to and be installed in accordance with DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19, 393.20, 393.22 through 393.31, 393.33, and as specified herein.

3.8.2.3 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Operating units shall be provided with a visible and audible flash indicator. Turn signals shall be of the self-canceling type.

3.8.2.4 Emergency flashers. Vehicle shall be provided with four-way emergency flashers with a visible and audible flash indicator. Emergency flashers shall be operable with ignition switch "ON" or "OFF".

3.8.2.5 Battery. The truck and gasoline engine/air compressor (see 3.11.1), shall be equipped with a 12V battery. The total reserve capacity rating shall be not less than 90 minutes. The total cold cranking ratings at 0 degrees (°) Fahrenheit (F) shall be not less than 380 amperes. Reserve capacity ratings and cold cranking ratings shall be measured in accordance with SAE J537. Batteries shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

3.9.3 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.65 and 393.67.

3.9.3.1 Air cleaner. The manufacturer's standard air cleaner shall be furnished.

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3.9.3.2 Fuel tank. Fuel tank shall be manufacturer's standard for the truck furnished with not less than 15 gallons total capacity.

3.9.4 Exhaust system. The vehicle exhaust system conforming to DoT Federal Motor Carrier Safety Regulations, section 393.83, shall be substantially constructed, effectively leak proof, and quiet in operation. Properly supported tailpipe shall extend to the rear bumper and shall be located to properly and effectively minimize the possibility to exhaust fumes entering the body. The exhaust system shall include muffler(s). The exhaust system beyond the engine compartment shall be insulated from the fuel and oil tanks and tank connections by securely attached metal shields at any point where it is 12 inches less from the fuel tank, oil tank connection, fuel line connections, or battery box.

3.9.5 Automatic transmission. Manufacturer's standard automatic transmission recommended for the make and model truck shall be provided. Transmission shall provide required performance under all normal operating conditions and under all conditions specified herein. Transmission shall include a torque converter and have not less than three forward speeds.

3.9.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.9.7 Frame. Chassis frame shall be manufacturer's standard for the truck furnished.

3.9.8 Suspension. Truck shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with truck loaded to rate gvw. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Truck shall be equipped with hydraulic, double-acting shock absorbers at the front and rear wheels.

3.9.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with truck loaded to rated gvw. A single reduction rear axle shall be provided.

3.9.10 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.40 through 393.42 and 393.45 through 393.48, 393.51, and 393.52.

3.9.10.1 Service brakes. Vehicle shall be equipped with power assisted, hydraulic actuated, four-wheel service brakes. The service brakes shall stop the vehicle, loaded to specified gvw, within the stopping distance requirements of DoT Federal Motor Carrier Safety Regulation 393.52.

3.9.10.2 Parking brake. Vehicle shall be equipped with a parking brake conforming to DoT Federal Motor Carrier Safety Regulations 393.41, and shall control and hold the truck, when loaded to its specified gvw on a 30 percent grade.

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3.9.11 Wheels, rims, tires. Truck shall be equipped with single front and rear wheels. Rim and tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on each truck. Disk type wheels shall be provided.

3.9.11.1 Tires. Tires shall be of tubeless type with highway tread. Tires shall be of rated capacity at a minimum equal to the load imposed on each tire, measured at each wheel, at the ground, with truck loaded to rated gvw.

3.9.11.2 Spare wheel. A spare wheel shall be mounted on a carrier and installed in a readily accessible location on the truck. Spare wheel shall be the same as mounted on the front and rear wheels (see 3.9.11).

3.9.12 Cab. Manufacturer shall furnish a standard full-width metal cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with at least the curbside door equipped with external, key-operated lock. Cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger seat. Interior dome lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be furnished and installed. Drip rails shall be installed above the cab doors. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.16). Light colored upholstery shall not be furnished.

3.9.13 Windshield wipers and washers. Truck shall be equipped with dual windshield wipers and washers. Windshield wipers shall be multispeed type and operated by electric motor(s).

3.9.14 Steering. When furnished, manufacturer's standard power steering shall be provided.

3.9.15 Heater and defroster. Hot water heater shall be provided. Heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.9.16 Rear end protection. Rear end of truck shall be protected in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.86. A rear bumper shall be provided.

3.9.17 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Key-operated ignition switch.
b. Ammeter, charging indicator or voltmeter.
c. Fuel gage.
d. Oil pressure gage or indicator.
e. Engine temperature gage or indicator.

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f. Speedometer with recording odometer.
g. Dual sunvisors.
h. Front door mounted armrest on driver's side.
i. Driver's compartment ventilator other than window.
j. Manufacturer's standard electric horn.

3.9.18 Outside rearview mirrors. Combination (flat and convex) outside rearview mirrors, enclosed in a common housing, shall be provided on each side of cab. Each combination mirror shall have a minimum 36 square inches of flat reflective area and an adjustable convex surface having a minimum 20 square inches of reflective area. Mirrors shall have not less than three supporting arms.

3.10 Weight and dimensions of Av-lube unit.

3.10.1 Weight. The Av-lube unit/truck shall not weigh more than 3,000 lbs when fully loaded and operational.

3.10.2 Dimensions. The dimensions of the Av-lube unit/truck shall conform to the inside dimensions of a fleet side pickup truck with the tailgate in the closed position. The Av-lube unit/truck shall be capable of sliding on its skid into and out of the pickup bed without interference with the wheel wells. Height shall be limited to 78 inches when mounted in the pickup truck bed with the oil tank empty.

3.11 Av-lube components. The Av-lube system shall consist as a minimum, the following components:

a. Gasoline engine.
b. Air compressor.
c. Air tanks(s) with safety valve.
d. Air pump.
e. Pressure relief valve.
f. 200 gallon tank.
g. Cabinet with doors to house machinery.
h. Electric immersion heater(s).
i. Suction hose and storage tube.
j. Oil filter.
k. Hose reel and hose.
l. Hand controlled servicing nozzle.
m. Meter, liquid service type.
n. Skid mounting and supports.

3.11.1 Gasoline engine/air compressor. Gasoline engine/air compressor shall supply a minimum of 4 cubic feet of air-per-minute at 100 pound-force per square inch (psi) and be capable of operating continuously for 4 hours, at any ambient temperature from +120°F to -20°F when serviced in accordance with the manufacturer's recommendations. The system shall have 12V electric starter/generator and battery (see 3.8.2.5). Ignition shall have radio suppression conforming to SAE J552. The unit shall be equipped with engine

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hour meter, ammeter and control located in the rear cabinet. Mounting of these units shall be in a sheltered location and accessible for maintenance. Electromagnetic radiation from the Av-lube unit/truck shall be within limits of SAE J551.

3.11.2 Air reservoir tank(s). Air reservoir tank(s) shall have a minimum capacity of 20 gallon liquid measure. The wet tank shall have 140 psi safety valve and a heated automatic tank drain.

3.11.3 Air pump. Air pump shall have a minimum of 4 gallons per minute oil conforming to MIL-L-22851 flow at 60°F at the nozzle and operate on an air pressure as low as 40 psi to a nozzle height of not less than 78 inches above the top of the tank. Pump shall be capable of pumping from the ground level into the top of the 200 gallon tank.

3.11.4 Oil pressure relief valve. Oil pressure relief valve shall be of the pressure type and set at 35 psi ±5 percent. The pressure relief valve shall be located in the bypass tank return line.

3.11.5 Oil tank. Oil tank shall be of a single compartment type, with a 200 gallon capacity, plus a minimum 3 percent expansion space. The design of the tank shall include provisions for an efficient heating system (see 3.11.14). The tank shall be constructed of 5,000 series alloy sheet aluminum, of sufficient thickness to provide complete structural integrity of the tank and meet performance requirements specified herein. The tank shall have a circular or semi-circular cross section with a service platform on top, for the full length of the Av-lube unit/truck. The flat top shall be reinforced, as necessary, since it will be used as a platform for oil servicing of air craft. The service platform shall be covered with an anti-skid material. The height of the platform should not be over 78 inches above the ground level when the Av-lube unit/truck is mounted in a pickup truck when empty. A bimetal temperature gauge, with not less than 2-1/2 inch dial, with a range of minus 40°F to 120°F, shall be mounted in the rear cabinet with the temperature sensing element extending into the rear of the tank.

3.11.6 Manhole. The tank shall have a top mounted circular manhole, with a minimum diameter of 10 inches. The manhole cover shall have 4 inch minimum water tight spring-type hold-down catch filler cover with necessary safety features to eliminate the possibility of a spark occurring during opening and closing of the cover. The cover, or tank, shall be equipped with a 2-way venting system and fusible plug to provide additional emergency pressure release. If the fill cover is hinged it shall be installed so that the hinge faces the front so that if the lid is left open, it immediately falls into position when the truck starts.

3.11.7 Service platform railing. A platform railing of a minimum 2-1/2 inches high, to provide protection against personnel slipping of the platform, shall be mounted on all edges of the top of the tank platform. The design of this guard shall be such that it will not trap water or spilled oil.

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3.11.8 Overturn protection. Tank top equipment, such as fill covers, manhole, and piping connections shall be provided with suitable protection in the event the tank overturns. Overturn protection shall not have closed ends or other construction features that will trap spilled product or water accumulation.

3.11.9 Rear cabinet. Rear cabinet shall be provided and constructed of aluminum so as to adequately support the equipment therein without undue vibration. In addition, the cabinet shall be so constructed as to be weather and splash proof and shall be provided with drains. The cabinet door shall be suitable stiffened and adequately supported, Powerise type counter balance, top hung, flush mounted, with continuous piano type stainless steel hinge and rod. Flush type hardware, including positive-type latches, chrome plated, shall be provided. An aluminum container, suitable for dry storage and labeled "Instruction Manuals", shall be attached to some accessible area inside the rear cabinet. The top of the cabinet shall be flat and shall be part of the service platform, and shall be suitably reinforced.

3.11.10 Pressure gage. A pressure gage, with a scale 0 to 150 psi, with not less than 2-1/2 inch dial, shall be provided and connected to the air tank and shall be located in rear cabinet. The gage shall incorporate a means to dampen pressure oscillation to afford stability of the indicating needle.

3.11.11 Oil piping. All oil piping shall be steel tubing with wall thickness to withstand 100 psi working pressure and 150 psi test pressure. The piping sizes shall be sufficient to meet the flow requirements of paragraph 3.11.3. All lines shall be as direct as possible with sweeping bends. A line stainer with 40 mesh screen shall be provided in drum unload pipe. The bypass tank return line shall enter the tank through the rear bulkhead. The drum unloading pipe shall be so located as to provide for draining the tank by gravity. A quick coupler, cap with chain, shall be provided on the end of drum unloading suction line.

3.11.12 Piping valves. All piping valves shall be of the gate type. All valves shall be readily accessible from the rear of the cabinet. A gate valve shall be provided at the bottom of the tank for emergency shut off.

3.11.13 Piping connections. Supports shall be provided where necessary to relieve piping strain, and all pipes and manifolds shall have joints where necessary, to facilitate disassembly for repair and maintenance. Piping arrangement shall allow for disassembly of all piping without removal of other components.

3.11.14 Heating system Unless otherwise specified in this specification, a heating system shall be provided for ease in pumping the aviation lube oil. One or more 110V immersion type heaters of adequate capacity, to maintain oil temperature of not less than 50°F in -20°F weather, will be accepted. Fifty feet of three wire cord with a commercial type twist lock plug at rated amperage shall be provided. No open flame type heaters will be permitted. A

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quick disconnect shall be provided where the electrical cord is attached to the Av-lube unit, the other end of the electrical cord shall have 2 pole, 3 wire grounding type plug attached and wired.

3.11.15 Drum suction hose and pipe. A drum unloading suction hose and pipe shall be provided with the provision for stowing in the rear cabinet with hold down clips. The hose shall be suction type, 8 feet long, one end shall have a quick coupler with dust plug and securing chain and the other end shall have a brass male coupling. The suction pipe shall be brass, 40 inches long, notched at the lower end with pipe thread at top. The suction hose and pipe unit shall be assembled.

3.11.16 Oil filter. A lubricating oil filter having a selectivity equal to a minimum of 150 microns shall be provided. No paper or material other than cellulose type material shall be used in the filter or the filter elements. The element shall be capable of withstanding a 50 psi differential pressure without rupture or mechanical failure. The filter will be installed upstream from the 35 psi pressure relief valve takeoff line.

3.11.17 Hose reel. A spring loaded, ratchet type, automatic rewind, hose reel shall be provided in the rear cabinet and shall have a capacity of 35 feet of hose.

3.11.18 Hose. Thirty feet of hose shall be provided. Hose shall accommodate aviation lubrication oil specified herein. The hose shall be adequate in size to meet the flow requirements in paragraph 3.11.3.

3.11.19 Nozzle. A hand controlled servicing type nozzle with a spring loaded cap shall be provided with a 1 inch maximum size external nozzle spout diameter.

3.11.19.1 Meter. An all liquid service type meter shall be provided, installed between the relief valve and hose reel inlet and shall read in gallons and quarters of gallons.

3.11.20 Platform ladder. A means for stepping from the pickup truck bed to the left rear of the platform tank shall be provided. Ladder shall be provided with suitable grab handles in order than personnel may ascend to and descend from the tank platform in a safe manner. Treads of ladder and walking surfaces of all walkways shall be of the antiskid type.

3.11.21 Lighting. One or more vaporproof and shatterproof lights shall be located in the rear cabinet to provide light for the cabinet interior so that instruments installed in the cabinet may be clearly read. Waterproof and vaporproof wiring shall be used exclusively. The voltage of all lamps shall be 12V. A single flood light with a 180° vertical and a 360° horizontal rotation, mounted on top rear center of the rear cabinet shall be provided. A two lens 360° rotational amber beacon light shall be mounted at the front center of the service platform, extending over the truck cab.

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3.11.22 Fire extinguisher brackets. Brackets for two 20-lb capacity dry chemical fire extinguishers conforming to A-A-393, type I, shall be installed in an outside sheltered location from which the extinguishers can be readily released in one operation with one hand when a person is standing on the ground.

3.11.23 Grounding. All metal parts of the Av-lube unit/truck shall be bonded together to prevent the possibility of a spark due to static electricity. A grounding cable between the Av-lube unit/truck and the pickup truck shall be provided.

3.11.24 Instruction plates. The Av-lube unit/truck shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws, rivets or bolts of not less than 1/8-inch diameter.

3.11.25 Piping diagram. An etched or embossed plate of corrosion-resistant material, showing piping diagram and instructions for operation, shall be attached to cabinet in a position for easy study.

3.11.26 Skid. A skid shall be provided of aluminum I beam construction with adequate flanges to support the weight and be resistant to rough handling. The skid shall be fastened to the truck frame in at a minimum 4 places, using "J", or similar type bolts.

3.12 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.13 Safety. All rotating or reciprocating parts and all parts subject to high operational temperatures that are of such a nature, or are so located, as to be or become a hazard to operating or attending personnel, shall be subtantially guarded or insulated to the extent necessary to eliminate the hazard. All ladder and walk ways external of the tank shall be skid proof. Ladder and hand holds shall be provided in such quantity and size on the Av-lube unit/truck entrance thereto and exit therefrom all be unhampered and non-hazardous.

3.14 Sound level. The Av-lube shall be provided with all standard and optional sound suppression equipment to reduce the exterior sound level to a maximum of 84 decibles (db(A)) at the operator's or any other occasionally occupied position. If the sound level exceeds 84 db(A) the contractor shall provide warning decals and appropriate warnings in manuals in accordance with MIL-STD-1474. The sound level shall not exceed 88 db(A).

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3.15 Identification marking and data plates. Identification marking and data plates shall be in accordance with MIL-STD-1223 as required for the delivery point user. Fuel tanks shall be clearly marked on or near the filler cap "Gasoline" in red block letters with a minimum of one inch in height (black letters for camouflage patterns).

3.16 Cleaning, treatment, and paintings. Cleaning, treatment, and painting shall be in accordance with MIL-STD-1223, as required for the Navy. Exterior color of the Av-lube unit/truck shall be yellow color No. 13538 in accordance with FED-STD-595. The interior cabinet surface shall be painted the same as the exterior surfaces.

3.17 Rustproofing. When specified (see 6.2), the truck cab, chassis and chassis underside shall be rustproofed in accordance with FED-STD-297.

3.18 Servicing and adjusting. Prior to acceptance of the Av-lube unit/truck by the Government, the contractor shall service and adjust the Av-lube unit/truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The Av-lube unit/truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.19 Workmanship.

3.19.1 Metal fabrication. The metal used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the metal to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.19.2 Fastening devices. Fastening devices shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.19.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not

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countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.19.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.19.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contaractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Governemnt for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production vehicle inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the test of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each Av-lube unit/truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First production vehicle test. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.4.1 Cleaning. Prior to the performance of any test, each oil dispensing system shall be carefully inspected for evidence of contaminants in the tank and pumping system, such as loose, spattered of excess weld or solder chips and foreign material. Each oil dispensing system shall then be thoroughly steamed cleaned and the entire system flushed with aviation lubricating oil then drained. Prior to test of shipment filter elements shall be installed.

4.4.2 Pressure checks. The oil tank of the Av-lube unit/truck shall be subjected to the internal static air pressure of 3 psi for a period of not less than 30 minutes. A soap solution shall be applied to the exterior of the tank and around the manhole and filler cover to facilitate detection of leakage. The oil pumping system from the tank gate valve on suction side of the oil pump to the nozzle, shall be subjected to a hydrostatic pressure of 35 psi pressure and the relief valve shall be subjected to a hydrostatic pressure of 35 psi for not less than 30 minutes. Evidence of leakage under any of the pressure tests designated above shall be cause for rejection.

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4.4.3 Av-lube unit weight and tank capacity. The skid mounted unit shall be weighed to determine conformance to 3.10.1 and the oil tank capacity shall be checked for conformance to 3.11.5.

4.4.4 Road tests. The truck, loaded to provide not less than rated gvw, shall be driven at speeds up to 50 mph over improved roads for not less than 30 miles, and at speeds of 20 mph over unimproved roads for not less than 20 miles. Upon completion of the test, the truck and equipment shall be examined for leaks, warping, distortion, or malfunction.

4.4.5 Parking brake. Parking brake will be tested to assure fully loaded vehicle will hold on a 30 percent grade to determine conformance to 3.9.10.2.

4.4.6 Measurement of electromagnetic radiation. To determine conformance to 3.5.3 and 3.11.1, electromagnetic radiation shall be measured in accordance with SAE J551. When suppressed to conform to 3.5.3 and 3.11.1, the manufacturer shall furnish certification, in lieu of tests that the Av-lube unit and truck each meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.4.7 Engine/air compressor. The engine/air compressor shall be operated to determin conformance with the requirements to 3.11.1.

4.4.8 Flow rate. Not less than 55 gallons of aircraft lubricating oil, conforming to MIL-L-22851, shall be pumped through the pumping system to produce flow rates conforming to the requirements of 3.11.3.

4.4.9 Drum unloading. The drum suction hose and pipe (see 3.11.15), shall be installed in accordance with the manufacturer's recommendations. The pumping system shall demonstrate its ability to pump oil from drum located on the ground level. A minimum of one 55 gallon drum of aircraft lubricating oil, conforming to MIL-L-22851, shall be pumped from the drum into the oil dispensing tank to determine conformance to 3.11.3 and 3.11.15.

4.4.10 Heating system. The heating system, when furnished, shall operate for a sufficient length of time to insure proper functioning of the system as specified in 3.11.14.

4.4.11 Sound level. Sound level test shall be performed to determine conformance to 3.14. In lieu of the test, the manufacturer may submit a certification that the truck meets the requirement.

4.4.12 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

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4.5 Production unit tests. The contractor's testing system shall, as a minimum, assure that the fully equipped Av-lube unit/truck is capable of meeting the performance requirements specified herein. Tank shall be subjected to the cleaning requirements of 4.4.1, and the pressure tests specified in 4.4.2.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The Av-lube unit/truck shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved, packaged, and marked in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.2).

6. NOTES

6.1 Intened use. The Av-lube unit/truck covered by this specification is intended for use at established Naval Stations for transporting and dispensing aviation lubricating oil to aircraft, both within and without the continental limits of the United States. They will be operated over improved and unimproved roads having grades usually found on main and access roads in and about established naval bases.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first production vehicle is required (see 3.2).
c. When rustproofing is required (see 3.17).
d. When preservation-packaging is in accordance with MIL-STD-281 is required and the level of preservation-packaging is required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.4 First production vehicle. When a first production vehicle inspection is required, the vehicle shall be tested and should be a first production vehicle or a standard vehicle from the contractor's current inventory. The contracting officer should include specific instructions in the contract regarding arrangements for examination, test, and approval of the first production vehicle (see 3.2 and 4.2.1).

Custodian:

Navy - YD

Review Activity:

Army - AT

Preparing Activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-29246	TRUCK, AIRCRAFT LUBRICATING OIL DISPENSING UNIT, 200 GALLON, SELF- CONTAINED, SKID MOUNTED, 4 BY 2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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